

Institute of Actuaries of India

Subject CM2A – Financial Engineering and Loss Reserving (Paper A)

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) Answer : IV

x_1 follows lognormal distribution, x_2 follows lognormal distribution,
then $x_1 * x_2$ follows lognormal

(2)

ii) Answer : I all of the above

(2)

iii) Answer III. The unexpected dividend will decrease the share price on ex-dividend date. As this is not anticipated the value of call option will reduce and put option will increase

(2)

iv) Answer : III only d

(2)

v) Answer III. b and c since a & d invalidate the assumptions of the theory

(2)

Solution 2:

i)

- I. weak form
- II. Semi strong
- III. strong form
- IV. strong form.

(2)

ii)

a)

- There is status quo bias here. [$\frac{1}{2}$]
- the team is sticking with the current situation [$\frac{1}{2}$]
- even in the face of a favourable alternative [$\frac{1}{2}$]
- The team are also showing herd behaviour [$\frac{1}{2}$]
- They are following/mimicking the actions of others in order to manage an uncertain situation [$\frac{1}{2}$]

(2)

b) This is an example of anchoring and adjustment. The customer's expectation with regard to future car loan interest rates are anchored by past experience of low interest rates. Consequently, they expect interest rates to revert back to average historical levels even when there is no evidence that they are likely to do so.

(2)

iii) IV. If Intrinsic value is positive the option is in-the-money and ₹10 can be the value of any put or call option as well

(2)

iv) Answer II

(3)

Using the standard symbols

calculate $d_1 = (\ln(104/100) + (0.12 + .3^2 / 2) * .25) / .3 * \text{sqrt}(.25)$

= .5365

$d_2 = .3865 (d_1 - .3 * \text{sqrt}(.25))$

Price of call option

$$c = 104 * \Phi(.5365) - 100e^{-.03} * \Phi(.3865) \text{ ----- (1)}$$

$$= 10.11$$

From equation (1) if the current share price and strike price is divided by 2, the price of call option will also be half of the calculated value.

[11 Marks]

Solution 3:

- i) Utility functions exhibiting **constant relative risk** aversion are said to be isoelastic Iso-elastic means that the elasticity of the marginal utility of wealth is constant with respect to wealth.

Comment :- Single line definition is fine. Highlighted words are critical

(1)

- ii) Only (III) does not exhibit iso-elasticity as its relative risk aversion measures are given by $R'(w) = -2d/(1+2dw)^2$ is not constant. (2)

iii)

- a) $U'(w) = 1+2dw$ and $U''(w) = 2d$, Because the contestant is risk averse, we must have $U''(w) < 0$, so we must have $d < 0$

The condition of non-satiation requires $U'(w) > 0$

Hence $1+2dw > 0$, and $w < -1/2d$

So the quadratic utility function can only satisfy the condition of non satiation over a limited range of w i.e $-\infty < w < -1/2d$

Now $10,00,000 = -1/2d$, so $d = -0.0000005$

Comment:

$$U'(W) = \frac{1}{2}$$

$$U''(W) = \frac{1}{2}$$

Condition $-1/2$

Quadratic utility satisfaction range $-1/2$

(3)

$$b) U(250,000) = 250,000 - 0.0000005 * (250,000)^2 = 218,750$$

$$E[U(\text{exchange})] = 0.5 \times U(600,000) + 0.5 \times U(0)$$

$$= 0.5 \times (600,000 - 0.0000005 \times 600000^2) = 210,000$$

So the contestant should not accept the opportunity to exchange because the expected utility of the exchange opportunity is lower than that of the prize.

(3)

[9 Marks]

Solution 4:

i)

- a) The Shortfall probability measures the **probability of returns falling** below certain chosen **benchmark**. (1)

b) VaR gives the maximum potential loss in the value of the portfolio with a given degree of confidence over a given future time period. (1)

ii) Answer : IV since variance of Weiner processes at and bt are t & 0.75 would out leaving only t (3)

iii) The single day volatility or standard deviation of daily change of both Asset A and B is $1000 \times 1\% = 10$

So the variance of daily change of the portfolio

$$= 10^2 + 10^2 + 2 \times 0.3 \times 10 \times 10$$

$$= 260$$

(1)

So the standard deviation of daily change of the portfolio is =

$$\text{SQRT}(260) = 16.125$$

(1)

So the standard deviation of 4-day change = 2×16.125

$$= 32.25$$

(1)

From the Tables we know that $\Phi(-2.33) = 0.01$

$$\text{So 4-day 99\% VaR} = 2.33 \times 32.25 = 75.14$$

(1)

[Max 4]

[9 Marks]

Solution 5:

i) The process of holding suitable derivatives and the underlying assets in appropriate combinations to eliminate the market risk facing the portfolio is known as hedging.

(1)

Arbitrage is generally described as a **risk-free trading profit**. Generally, an arbitrage opportunity exists if either an investor can make a deal that would give an immediate profit with no risk of future loss or an investor can make a deal that has **zero initial cost and no risk of future loss with a non-zero probability of a future profit**.

(1)

[Max 2]

ii) Put-call parity for a non-dividend paying share

$$p + S = c + Ke^{-r(T-t)}$$

a) Differentiating with respect to S

$$\partial p / \partial S + 1 = \partial c / \partial S$$

Hence the delta of call is 1 higher than delta of put. Option ii) is correct

(1)

b) Differentiating with respect to σ

$$\partial p / \partial \sigma = \partial c / \partial \sigma$$

Shows Vega of European put is equal to European Call. Option i) is correct

(1)

c) Differentiate with respect to S again from a)

$$\partial^2 p / \partial S^2 = \partial^2 c / \partial S^2$$

Gamma of European call and put are equal. Option iii) is correct

(1)

iii) The lower bound of the put option

$$p_t \geq \text{Max}(Ke^{-r(T-t)} - S_t, 0)$$

in this case

the present value of the strike price $90e^{-.10 \cdot 1/2} = 85.61$

and since

$5 < 85.61 - 80$ which does not adhere to the lower bound condition. (1)

Hence arbitrage opportunity exists and it can be utilized as follows

Borrow ₹ 85.61 at 5% and buy one share and put option on that share which costs 85. (1)

If the share price is above 90 at expiry of term the put option is worthless but the share can be sold for 90. This strategy has generated profit which has a present value of at-least ₹0.61 (1)

If the share price is below 90 the put option can be exercised and the share can be sold for 90. In this case also it generates a profit with present value of at-least ₹0.61. (1)

[Max 4]

[9 Marks]

Solution 6:

- i) a) is false since there cannot be a claim until time 2 (1)
 b) is false since the insurance company could be ruined at time 3 if there is a claim, if U is sufficiently small. (1)
 c) is false since the insurance company cannot be ruined in year 4, since by that stage it will have sufficient premiums to cover any loss. (1)
 d) is true since if it is not ruined by time 4, the insurance company cannot be ruined (1)

ii) The premium charged will be:

$$1.3 \cdot 0.2 \cdot (0.25 \cdot 10 \text{ Crores} + 0.75 \cdot 1 \text{ Crore}) = \text{Rs. } 0.845 \text{ Crores} \quad (2)$$

iii) The possibilities are tabulated below, where N means not injured, R means injured but recovered and X means injured but career ending:

Year 1	Year 2	Probability	Ruin	Marks
N	N	$0.8 \cdot 0.8 = 0.64$	No	
N	R	$0.8 \cdot 0.2 \cdot 0.75 = 0.16$	No	
N	X	$0.8 \cdot 0.2 \cdot 0.25 = 0.04$	Yes	1
R	N	$0.2 \cdot 0.75 \cdot 0.8 = 0.16$	No	
R	R	$0.2 \cdot 0.75 \cdot 0.2 \cdot 0.75 = 0.0225$	No	
R	X	$0.2 \cdot 0.75 \cdot 0.2 \cdot 0.25 = 0.0075$	Yes	1
X	N/A	$0.2 \cdot 0.25 = 0.05$	Yes	1

Summing the cases where ruin occurs we have

$$\psi(1 \text{ Crore}, 2) = 0.04 + 0.0075 + 0.0025 = 0.0975 \quad (1)$$

[4]

[10 Marks]

Solution 7:

Bond holders would receive : 3 crores ;
if $A(5) - 6.04 * 1.03^5 \text{ cr} \geq 3 \text{ crores}$

where $A(5)$ is the value of Asset at time 5

And

$A(5) - 6.04 * 1.03^5 \text{ cr}$;

if $A(5) - 6.04 * 1.03^5 \text{ cr} < 3 \text{ crores}$

Bond holders would receive the $\text{Min}(3 \text{ crores}, A(5) - 6.04 * 1.03^5 \text{ cr})$ (1)

Let $L2(5) = 3 \text{ crores}$, $L1(5) = 6.04 * 1.03^5 \text{ cr} = 7 \text{ crores}$

Bond holders would receive $\text{Min}(L2(5) , \max(A(5) - L1(5), 0))$ (1)

Or

$L2(5) - \max(L2(5) - \max\{A(5) - L1(5), 0\}, 0)$

$=> L2(5) - \max(L2(5) + L1(5) - A(5), 0)$ (1)

(max 3 marks for arriving at the correct format)

The second part is a put option with strike price $L2(5) + L1(5) = 10 \text{ crores}$

Put option value at time 0:

$K = 10$, $S_0 = 15$, $e^{(.05*5)} = .778801$, ; $d1 = 1.45204$, $d2 = 0.89302$ (2)

$\phi(-d2) = 0.185922$, $\phi(-d1) = 0.073245$

Using the above in Black Scholes formula for put option

$= p = 0.349$ (1)

Value of the bond at time 0:

$3 * e^{(.05*5)} - 0.349$ (0.5)

$=> 3 * 0.778801 - 0.349 = 1.987 \text{ crores}$ (0.5)

[7 Marks]

Solution 8:

i) (c) is the correct answer (2)

ii)

Underwriting year	Development year				Marks
	0	1	2	Premium Income	
2017	38,420	78,110	91,000	1,20,400	
2018				1,17,100	

	31,500	79,000			
2019	44,000			1,35,500	
Column total		1,57,110	91,000		0.5
Column total excluding last	69,920	78,110			
Dev factor	2.2470	1.1650			
f	2.6178	1.1650			0.5
Underwriting year	Premium	Initial UL			
2018	1,17,100	105390			1
2019	1,35,500	121950			1
	Initial UL	f	1-1/f	Emerging Liability	
2018	1,05,390	1.1650	0.1416	14,928	0.5
2019	1,21,950	2.6178	0.6180	75,365	0.5
				90,293	1

The estimate of outstanding claim is Rs.90,293

(5)

iii) Assumptions:

1. Payment pattern same for each underwriting year.
2. Estimated Loss Ratio is appropriate.
3. Claims from underwriting year 2017 are fully run off.
4. Data already adjusted for inflation or past pattern of inflation will be repeated in future.

(2)

[9 Marks]

Solution 9:

Find the rate corresponding to probability 50%

$$.0735 = .05 \cdot .2 + .07 \cdot .3 + x \cdot .5$$

(1)

$$x = .085$$

(0.5)

Expected accumulated value from the Government scheme = $((1.05)^{10} \cdot .2 + (1.07)^{10} \cdot .3 + (1.085)^{10} \cdot .5) \cdot 2 \text{ lakh} = 4,09,283.21$

(1)

Payment to the Bank:

co-efficient of skewness = $2/3$, using the formula in page 12 of "formula tables", we get $\alpha = 9$

(1)

Interest charged follows gamma with expected return of .065

(0.5)

$$= > .065 = 9 / \lambda \Rightarrow \lambda = 138.46$$

(1)

$$\text{Variance[interest charged]} = \alpha / \lambda^2 = 0.00047$$

(1)

Let A_{10} be the accumulated value of amount payable

To find : $P \{ A_{10} > 409283 \}$

(1)

$$E [A_{10}] = 2 \text{ lakh} \cdot 1.065^{10} = 3,75,427.49$$

(1)

$$V[A10] = 2 \text{ lakh}^2 * ([1.065^2 + .00047]^10 - 1.065^20) = 584,461,380 \quad (2)$$

Using normal approximation;

$$\begin{aligned} &P\{(A10 - \text{mean}) / \text{sd} > 409283\} \\ \Rightarrow &1 - P\{(A10 - \text{mean}) / \text{sd} < (409283.21 - 3,75,427.49) / 24175.64\} \quad (1) \\ \Rightarrow &1 - P\{Z < 1.40\} \quad (0.5) \\ \Rightarrow &8.08\% \quad (0.5) \end{aligned}$$

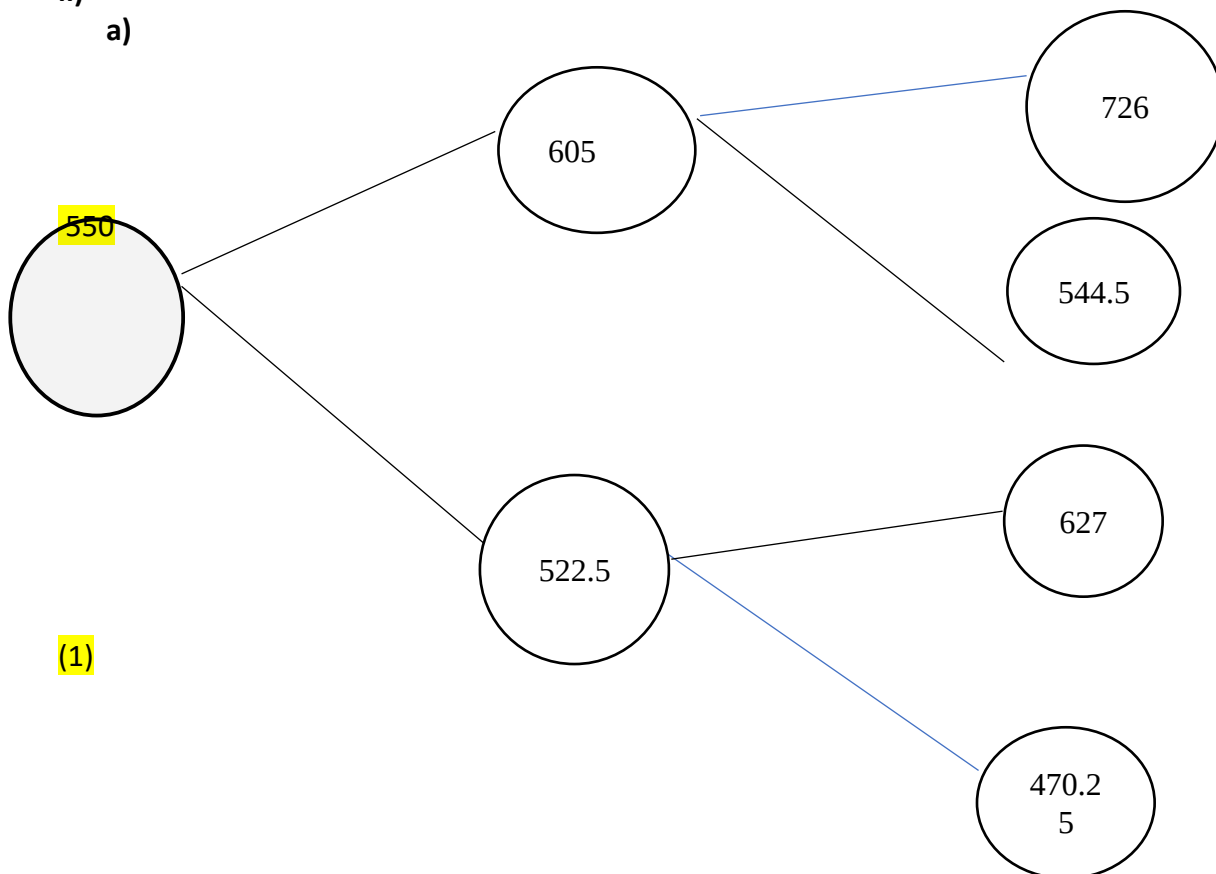
[12 Marks]

Solution 10:

- i) The recombining binomial tree is assumed to have the size of up-steps and down-steps to be same under all states and across all time intervals. It means $u_t(j)=u$ and $d_t(t)=d$ for all times t and states j , also $d < e^r < u$ (1)
Hence the risk neutral probability 'q' is also constant for all times and in all states. (1)
The main advantage of recombining binomial tree is that it greatly reduces the computational time by reducing the number of states. (1)
[3]

ii)

a)



The risk-neutral probability for each state

$$\begin{aligned} q_1 &= (e^{0.02} - 0.95) / (1.10 - 0.95) = 0.4680 \text{ (time 1)} \\ q_2 &= (e^{0.03} - 0.90) / (1.20 - 0.90) = 0.4349 \text{ (time 2)} \end{aligned}$$

(1)

Possible Payoffs after six months are

$$p_{uu} = \text{Max}(500-726,0) = 0$$

$$p_{ud} = \text{Max}(500-544.5,0) = 0$$

$$p_{du} = \text{Max}(500-628,0) = 0$$

$$p_{dd} = \text{Max}(500-470.25) = 29.75$$

(1)

The value of the option at first 3 months if upside

$$V_1(1) = e^{-0.03} (0.4348 * 0 + 0.5652 * 0) \\ = 0$$

For downside

$$V_1(2) = e^{-0.03} (0.4348 * 0 + 0.5652 * 29.75) \\ = 16.3178$$

(2)

At time 0 the value of the put option will be

$$V_0 = e^{-0.02} (0.4680 * 0 + 0.5320 * 16.3178) \\ = 8.5092$$

(1)

[Max 6]

b) For the first 3 months, the payoff if upside

$$\text{Max}(500-3/6*605,0) = 197.5$$

And if downside

$$\text{Max}(500 - 3/6*522.5,0) = 238.75$$

(1)

The value of the option at first 3 months [since the final payoffs are same as ii)a)]

$$V_1(1) = 0 \text{ and } V_1(2) = 16.3178$$

It is optimal to exercise the American put option early in this case as the payoffs are high compared to the value of put option at first 3 months

(1)

Hence, the value of the put option assuming it is exercised at the first 3 months

$$V_0 = e^{-0.02} (0.4680 * (197.5-0) + 0.5320 * (238.75-16.3178)) \\ = 206.5905$$

(2)

At expiry the American put option value is same as the European put option derived in ii) a) which is = 8.5092

Hence, the total value of American put is $206.5905 + 8.5092 = 215.0997$

(1)

[Max 5]**[14 Marks]**
